

CB23F35: Low-Code Application Development using Zoho Creator

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Ex. 1: Comparative Case Study of Traditional Development and Low-Code Development Using Zoho Creator

AIM:

To perform a comparative case study of traditional application development and low-code application development using Zoho Creator, and to analyze their differences in terms of development time, coding requirements, ease of use, customization, maintenance, and overall efficiency.

OBJECTIVE:

The objectives of this experiment are:

1. To understand the traditional approach to application development.
2. To understand the low-code development approach using Zoho Creator.
3. To develop a simple application using both approaches.
4. To compare the two approaches based on development effort, time, flexibility, and maintenance.
5. To identify suitable use cases for traditional and low-code development.

SOFTWARE REQUIREMENTS:

Requirement	Specification
Traditional Development	HTML, CSS, JavaScript / Python / Java
Low-Code Platform	Zoho Creator
Database	MySQL / PostgreSQL or equivalent
Browser	Google Chrome / Microsoft Edge
Operating System	Windows 10/11
Internet	Required for Zoho Creator

THEORY:

Traditional Application Development

Traditional application development involves manually writing program code to create the user interface, application logic, database connectivity, validation, and other functionalities.

The developer generally needs to design the database, create forms and pages, write backend logic, establish database connections, implement validation, and test the application manually.

Low-Code Application Development

Low-code development allows applications to be created using visual interfaces, drag-and-drop components, predefined templates, workflows, and configuration-based features with minimal manual coding.

Zoho Creator is a low-code application development platform that allows users to create forms, reports, pages, workflows, and applications using a visual development environment. Deluge scripting can be used when additional customization or logic is required.

CASE STUDY:

For comparison, a simple Student Management Application can be considered. The application contains the following modules:

- Student Profile
- Course Enrollment
- Academic Record
- Guardian Contact

The same basic application functionality is considered for both development approaches.

Traditional Approach

In the traditional approach, the developer manually creates:

- HTML forms for entering student details
- CSS for designing the interface
- JavaScript for form validation
- Backend code for processing data
- Database tables for storing records
- SQL queries for inserting and retrieving records

- Authentication and access control
- CRUD operations for managing student information

Zoho Creator Low-Code Approach

In Zoho Creator:

1. A new application is created.
2. Forms are created using the visual form builder.
3. Fields such as Student ID, Name, Department, Email, and Phone Number are added.
4. Reports are generated from the forms.
5. Workflows are configured for required actions.
6. Validation rules can be added through configuration or Deluge scripts.
7. The application can be tested and deployed from the platform.

PROCEDURE:

A. Traditional Development

1. Define the requirements of the Student Management Application.
2. Design the database structure.
3. Create the required database tables.
4. Develop the frontend using HTML and CSS.
5. Implement validation using JavaScript.
6. Develop backend functionality.
7. Connect the application with the database.
8. Implement CRUD operations.
9. Test each module.
10. Deploy the completed application.

B. Low-Code Development Using Zoho Creator

1. Log in to Zoho Creator.
2. Create a new application.
3. Create the required forms using the drag-and-drop form builder.
4. Add appropriate fields and field types.
5. Configure mandatory fields and validation rules.
6. Create reports for displaying stored records.
7. Configure workflows for required automation.
8. Add Deluge scripts wherever additional logic is required.
9. Test the application.
10. Deploy the application.

COMPARATIVE ANALYSIS:

Parameter	Traditional Development	Zoho Creator Low-Code
Coding Requirement	High	Low
Development Time	Relatively high	Relatively low
Technical Skill	High	Moderate
UI Development	Manually coded	Visual builder
Database Setup	Manually designed	Mostly platform-managed
Validation	Programmed manually	Configuration + scripting
Workflow	Manually implemented	Visual workflows
Customization	Very high	High, within platform capabilities
Maintenance	Requires developer effort	Comparatively easier
Deployment	Requires configuration	Platform-supported deployment
Learning Curve	Higher	Lower
Development Cost	Can be higher	Can be lower for simple applications
Scalability	Highly customizable	Depends on platform capabilities
Best suited for	Complex/custom systems	Business applications and rapid development

ADVANTAGES AND DISADVANTAGES:

Traditional Development

Advantages:

- Complete control over application architecture.
- High degree of customization.

- Suitable for complex applications.
- Greater flexibility in technology selection.

Disadvantages:

- Requires more programming knowledge.
- Longer development time.
- More effort for testing and maintenance.
- Database and deployment setup require additional work.

Zoho Creator Low-Code

Advantages:

- Faster application development.
- Requires less manual coding.
- Provides ready-made UI components.
- Simplifies database and form creation.
- Supports workflow automation.
- Suitable for rapid prototyping and business applications.

Disadvantages:

- Customization may be limited compared with fully coded applications.
- Depends on the platform's features and ecosystem.
- Advanced requirements may still require scripting.
- Platform-specific limitations may affect complex applications.

RESULT:

The comparative case study shows that traditional development provides greater flexibility and control but requires more programming effort and development time, whereas Zoho Creator's low-code approach enables faster application development with significantly less manual coding.